Roll No. | | | | | | | | | | | Total No. of Pages : 03

Total No. of Questions : 13

B.Pharmacy (Sem.-8)

BIOSTATISTICS AND RESEARCH METHODOLOGY

Subject Code : BP-801T

M.Code : 79764

Date of Examination : 02-01-2025

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Answer briefly :**

   a) What do you mean by Biostatistics?

   b) What are ideal characteristics of Measures of Dispersion?

   c) Define Null Hypothesis and Alternate Hypothesis.

   d) What is Probability?

   e) Distinguish between Parametric and Non-parametric tests.

   f) What is the need for Design of Experiments?

   g) Draw a pie-chart to represent percentage of herbal drug sector in different countries given in the following table:

Country	India	China	Germany	Italy	The Netherland	Japan	America	UK	Total
Percentage of herbal drug sector	13	18	24	11	5	14	10	5	100

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h) What are the benefits of MINITAB?

i) What is Factorial Design?

j) What is a Cohort Study? Give example.

SECTION-B

2. A geneticist recorded the following measurements taken on two week old mice of a particular strain. Is there evidence that the variance among mice in different litters is larger than one would expect on the basis of the variability found within each litter?

Litters						
I	II	III	IV	V	VI	VII
19.49	22.94	23.06	15.90	16.72	20.00	21.52
20.62	22.15	20.05	21.48	19.22	19.79	20.37
19.51	19.16	21.47	22.48	26.62	21.15	21.93
18.09	20.98	14.90	18.79	20.74	14.88	20.14
22.75	23.13	19.72	19.70	21.82	19.79	22.28

3. a) The following table shows the age distribution of cases of a certain disease reported during a year in a particular state.

Age	Number of cases
5-14	5
15-24	10
25-34	20
35-44	22
45-54	13
55-64	5

Compute the Mean, Median and Standard Deviation.

b) Discuss the method of Least Squares for Curve Fitting.

4. Explain in detail various steps involved in Report Writing.

SECTION-C

5. Write a detailed note on 'Multiple Correlation Analysis'.

6. Describe different methods of Sampling.

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7. The systolic blood pressures of 12 women between the ages of 20 and 35 were measured before and after administration of a newly developed oral contraceptive. Compute for statistical significance. 5

Subject	Systolic Blood Pressure (mmHg)	
	Before	After
1	122	127
2	126	128
3	132	140
4	120	119
5	142	145
6	130	130
7	142	148
8	137	135
9	128	129
10	132	137
11	128	128
12	129	133

8. Write a note on ' Kruskal Wallis test'.

9. Define the term Plagiarism. Why do we need to check Plagiarism?

10. Explain Confounding System for two level factorials.

11. Give key features and applications of MS-Excel.

12. Write a note on 'Central Composite Design'.

13. What are Experimental Design Techniques? Write in brief about its types.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

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